

Week 3 Master Guide

I have extracted every ounce of theory from the Instructor Deck , Week-3.pdf slides, merged the spin-offs from the v2 and 2023 assignments, pulled the exact formulas from your formula sheet.md , and translated the toughest concepts into “Explain Like I’m 5” analogies.

EVERY Week 3 Concept (Explained Like You’re 5)

THE FINAL EXAM TRAP LIST (Review Before Exam)

- **The Food Web Trap:** A **Food Chain** is a single, linear pathway of energy flow. A **Food Web** is a complex, interconnected network of multiple food chains. Exams often try to trick you by swapping these definitions.
- **The Stoichiometry Trap:** When calculating CO_2 emissions from combustion, always balance the chemical equation first. You cannot just use mass ratios without knowing the molar ratio (e.g., 1 mole of Methane produces 1 mole of CO_2 , but 1 mole of Butane produces 4 moles of CO_2).
- **The Arithmetic Population Trap:** If a question asks for population forecasting using decadal data but doesn’t specify a method, default to the **Arithmetic Increase Method** ($P_n = P_0 + n\bar{x}$) if the growth amount is relatively constant.

Part 1: Ecology (The Russian Nesting Dolls)

- **1. The Hierarchy of Life:** Think of nature like Russian nesting dolls. It starts tiny with an **Atom** → **Molecule** → **Cell** → **Organism** (one zebra) → **Population** (a herd of zebras) → **Community** (zebras, lions, and trees) → **Ecosystem** (the animals PLUS the non-living things like rocks, rain, and dirt) → **Landscape** → **Biosphere** (the whole Earth).
- **2. Biotic vs. Abiotic:** **Biotic** = Living things (bugs, birds, bacteria). **Abiotic** = Non-living things (sunlight, oxygen, temperature).
- **3. How Organisms Eat:** * **Producers (Autotrophs):** Plants and algae that make their own food using sunlight.
 - **Consumers & Decomposers (Heterotrophs):** Animals and fungi that have to eat *other* things to survive.

Part 2: Energy and The 10% Pizza Tax

- **4. The 10% Rule & Thermodynamics:** Imagine a pizza delivery guy who eats 90% of the pizza before handing it to the next guy. That is the **2nd Law of Thermodynamics** ("Entropy Rules!"). When a bunny eats grass, it burns 90% of the energy just running around and staying warm. Only **10%** of the energy is passed to the snake that eats the bunny.
 - *The Result:* This is why **top carnivores** (like tigers or eagles) are so rare! There is barely any energy left at the top of the pyramid.
- **Ecological Efficiency:** Explicitly defined as the percentage of usable energy transferred as biomass from one trophic level to the next (typically hovering around 10%).
- **Socio-Economic Pathways to Sustainability:** According to the *Global Biodiversity Outlook 4* (UN IPBES Report), the three most critical pathways to achieving global sustainability are:
 1. Reducing overall consumption.
 2. Reducing nitrogen emissions (agricultural runoff).
 3. Increasing overall agricultural productivity (growing more food on less land).
- **5. Ecological Pyramids:**
 - **Energy Pyramid:** ALWAYS upright. Energy is always lost as heat.
 - **Biomass Pyramid (THE TRAP):** Usually upright, but in the ocean, it can be **INVERTED!** Why? Phytoplankton (producers) are tiny and get eaten instantly, so their *standing mass* is small, but they reproduce so fast (high turnover) that they can support giant whales.
- **6. Biomagnification:** Imagine rolling a snowball in sticky glitter. As the snowball gets bigger, it picks up more and more glitter. Pollutants like **DDT, heavy metals, and POPs** (Persistent Organic Pollutants) dissolve in fat, not water. Because they don't break down, they accumulate and become millions of times more concentrated at the top of the food chain.

Part 3: The Rules & Population

- **7. Barry Commoner's 4 Laws of Ecology:**
 1. Everything is connected to everything else.
 2. Everything must go somewhere (like DDT ending up in eagles).
 3. Nature knows best.

- 4. **There is NO such thing as a free lunch** (every action has an environmental cost).
- **8. Population Dynamics:** Populations grow exponentially (like yeast making beer), following the formula $P = P_0 e^{rt}$.
 - *The Trap:* Declining birth rates do *not* lower the median age. Because people are living longer thanks to medicine, declining fertility creates an “**aging population**” where the median age goes UP.

Part 4: Environmental Chemistry (The Slide Secrets)

- **9. Activity vs. Concentration:** In super salty water (like the ocean), molecules crowd each other and don't act normally. We use an “Activity Coefficient” (γ) to measure this.
 - *The Trap:* Activity coefficients deviate the most from 1.0 when the water is super salty (**High Ionic Strength**) and the ions have a lot of charge (**High Valence**).
 - **10. Kinetics & Half-Lives:** Imagine cutting a cake in half. A **First-Order** half-life ($t_{1/2} = 0.693/k$) is the time it takes for half the chemicals to disappear.
 - *The Trap:* For first-order reactions, the half-life **never changes**, even if you double the starting amount of chemicals!
 - **11. Acid-Base (pH):** pH is the measure of Hydrogen ions. A pH of 7 is neutral. If pH drops, the water is acidic. *Formula:* $[H^+] = 10^{-pH}$.
 - **12. Thermodynamics (Gibbs Free Energy):** $\Delta G = \Delta H - T \Delta S$. For a reaction to happen spontaneously, ΔG must be negative.
 - *The Trap:* If a reaction releases heat (Exothermic, $\Delta H < 0$) but WON'T happen at cold temperatures, it means the entropy (ΔS) must be less than zero.
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100% COMPLETE Week 3 Master Q&A Bank

Ecology & Ecosystems

- **Question:** The part of the earth and its atmosphere in which organisms live is called the:
 - **Answer:** (b) Biosphere.
- **Question:** Combination of a group of species at a common place is called a:
 - **Answer:** (b) Community.
- **Question:** Which of the following best distinguishes an ecosystem from a community?
 - **Answer:** (D) Ecosystem includes biotic and abiotic interactions.
- **Question:** Biotic environment includes:

- **Answer:** Flora and Fauna (Living things like plants and animals).
- **Question:** The importance of an ecosystem lies in:
 - **Answer:** Both cycling of material AND flow of energy.
- **Question:** What is the logical sequence of energy flow?
 - **Answer:** Producer → Consumer → Decomposer.
- **Q: True or False: In any energy conversion, some energy is always lost as heat.** [2023]
 - **A:** True (This is the 2nd Law of Thermodynamics).
- **Q: What distinguishes “Ecosystem functions” from “Ecosystem structures”?** [v2 Guide]
 - **A:** Structures are the pieces (Biotic/Abiotic); Functions are the processes (**Energy flow** and **Nutrient cycling**).
- **Question:** Which of the following animals is a carnivore (tertiary consumer)?
 - **Answer:** (d) Tiger.
- **Q:** What are the key socio-economic pathways to sustainability cited by the UN IPBES Report?
 - **A:** Reducing consumption, reducing nitrogen emissions, and increasing agricultural productivity.
- **Q:** Which ecological term describes the percentage of usable energy transferred as biomass from one trophic level to the next?
 - **A:** Ecological Efficiency.
- **Q:** What is the difference between a population and a community? **A:** Population = group of individuals of the same species in an area; Community = group of different species in the same area.
- **Q:** What does an ecosystem include that a community does not? **A:** An ecosystem includes both the biotic community (living) AND the abiotic (non-living) environment.
- **Q:** Is soil biotic or abiotic? **A:** Abiotic, as it is a non-living or physical component of the environment.
- **Q:** What are producers classified as in ecological terms? **A:** Autotrophs (e.g., plants, algae, bacteria).
- **Q:** How much energy reaches the top trophic level from an initial 1000 Cal? **A:** Only 1 Cal (or 1 J from 1000 J).
- **Q:** Why is only 10% of energy transferred between trophic levels? **A:** Energy is used up for daily life activities, not all organisms that die are eaten, and not all parts of an organism are digested.
- **Q:** What is ecological efficiency? **A:** It describes the efficiency with which energy is transferred from one trophic level to the next.
- **Q:** What exactly are decomposers? **A:** Heterotrophs that break down dead organic matter; they cannot produce their own food directly from sunlight.

- **Q:** What is the key difference between energy flow and material cycling? **A:** Energy flow is unidirectional (cannot be recycled due to the 2nd Law), whereas matter cycles.
- **Q:** State the two laws of thermodynamics. **A:** 1st Law = energy changes forms but is not created/destroyed; 2nd Law = amount of usable energy decreases as energy changes forms ("Entropy Rules!").
- **Q:** What is the relationship between "Nature knows best" and the 2nd Law of Thermodynamics? **A:** 2nd Law states all systems tend toward minimum free energy; nature's evolved processes are optimal, and human-added disturbances carry hidden costs.
- **Q:** Which ecological pyramid is always upright and never inverted? **A:** Pyramid of energy.
- **Q:** Under what condition can a pyramid of biomass be inverted? **A:** If the lower trophic level has a high rate of turnover of small individuals (e.g., phytoplankton).
- **Q:** What are all 6 population forecasting methods? **A:** (i) Arithmetical increase, (ii) Geometrical increase, (iii) Incremental increase, (iv) Simple graph, (v) Decrease rate of growth, (vi) Comparative graph and master plan.
- **Q:** What four components determine growth rate r ? **A:** $r = (b - d) + (i - e)$ (births, deaths, immigration, emigration).
- **Q:** What are the forms and units of energy? **A:** Chemical, radiant, thermal, mechanical, nuclear, electrical. Units are Kilojoules (kJ) and Kilocalories (kcal).
- **Q:** What do population doubling times show about development levels? **A:** Regions of lowest development show the shortest time before population doubling.

Thermodynamics, Energy Pyramids & Biomagnification

- **Question:** A pyramid showing the number of organisms at each trophic level is called:
 - **Answer:** (a) Pyramid of number.
- **Question:** In an aquatic ecosystem, an inverted biomass pyramid is observed. Which combination of conditions makes this possible without violating thermodynamic laws?
 - **Answer:** (B) Low producer biomass with very high turnover rate.
- **Question:** Which ecological principle best explains why top carnivores are rare and vulnerable to extinction?
 - **Answer:** (A) Second law of thermodynamics (Because 90% of energy is lost as heat at each trophic level).
- **Question:** Phytoplankton gets 100 J of energy. It is eaten by Small Fish, which is eaten by Big Fish. What is the total energy lost before reaching the Big Fish?
 - **Answer:** (d) 99 J.

- *Note: Phytoplankton (100J) → Small Fish (gets 10J, loses 90J) → Big Fish (gets 1J, loses 9J). Total lost = 90 + 9 = 99J.*
- **Question:** Assertion: Biomagnification does not violate the Second Law of Thermodynamics. Reason: Energy decreases between trophic levels, but matter can accumulate.
 - **Answer:** (A) Both true, reason explains assertion.
- **Question:** Why are persistent organic pollutants (POPs) especially harmful at higher trophic levels?
 - **Answer:** (c) High lipid solubility and low metabolic breakdown.
- **Q:** Find the heat of combustion when methane is burned.
 - **A:** -802.2 kJ/mol of CH_4 .
 - *Note: Negative means the reaction is exothermic (releases heat).*
- **Q: What is Henry's Law and what does it measure?** [Slide Veto]
 - **A:** It describes the solubility of a gas in a liquid. Formula: $C_{\text{aq}} = K_{\text{H}} \times P_{\text{g}}$ (where C_{aq} is the concentration in water, P_{g} is the partial pressure of the gas, and K_{H} is the Henry's Law constant).
- **Q: Define the Arrhenius Equation and its significance.** [Slide Veto / Formula Sheet]
 - **A:** $k = Ae^{-E_a/RT}$. It shows how the reaction rate constant (k) increases exponentially with temperature and decreases with activation energy (E_a).
- **Q: What is the "Efficiency" of an energy conversion process?** [Slide Veto]
 - **A:** $\eta = \frac{\text{Useful Energy Output}}{\text{Total Energy Input}} \times 100$.
- **Q: Which law of thermodynamics defines "Entropy" and the direction of time?** [2023 Spin-off]
 - **A:** The **Second Law of Thermodynamics** (Energy tends to degrade into a less useful, more disordered form — heat).

Population & Ecological Laws

- **Question:** Which of the following is NOT a rule of ecology by Barry Commoner?
 - **Answer:** (d) "There is always a thing as a free lunch."
 - *Note: The actual 4th law is "There is NO such thing as a free lunch." This is a classic trick question.*
- **Question:** Which of the following statements about population dynamics is FALSE?
 - **Answer:** (B) Declining fertility always reduces median age.
 - *Note: Declining fertility actually leads to population aging, which increases the median age.*

- **Question:** Find population in 2030 by arithmetic increase method. Data: 1980=10000, 1990=13000, 2000=18000, 2010=23000.
 - **Answer:** (d) 31,667.
 - **Note:** $P_n = P + n \cdot c$. Increments are 3000, 5000, 5000. Average increment $c = 4333.3$. For 2 decades (2010 to 2030): $23000 + 2(4333.3) \approx 31667$.
- **Q:** Which of the following are pathways to sustainability using the socio-economic scenarios?
 - **A:** All of the above (Climate mitigation, Improve agricultural productivity, Reduce waste).

Environmental Chemistry & Kinetics

- **Question:** Activity coefficients deviate most significantly from unity (1.0) when:
 - **Answer:** (B) Ionic strength is high and ions have high valence.
- **Question:** A reaction is exothermic but non-spontaneous at low temperature. Which condition must be true?
 - **Answer:** (A) $\Delta H < 0$ and $\Delta S < 0$.
- **Question:** For a first-order decay process, doubling the initial concentration will:
 - **Answer:** (c) Not change the half-life.
 - **Note:** $t_{1/2} = \frac{\ln(2)}{k}$, which has no concentration variable.
- **Question:** In a first-order reaction, find the relation between $t_{99\%}$ and $t_{90\%}$.
 - **Answer:** (d) $t_{99\%} = 2 \times t_{90\%}$.
- **Question:** Half-lives: Xylene = 56 days, Hexane = 15 days. Find their first-order rate constants.
 - **Answer:** (d) 0.012/day for Xylene, 0.046/day for Hexane.
 - **Note:** $k = 0.693/56 = 0.012$; $k = 0.693/15 = 0.046$.
- **Question:** Calculate CO_2 released by burning 32 g of methane completely.
 - **Answer:** (c) 88 g.
 - **Note:** $\text{CH}_4 + 2\text{O}_2 \rightarrow \text{CO}_2 + 2\text{H}_2\text{O}$. MW of Methane = 16. 32g = 2 moles. 2 moles of CH_4 produce 2 moles of CO_2 . $2 \times 44\text{g} = 88\text{g}$.
- **Question:** Find the hydrogen ion concentration and the hydroxide ions concentration in tomato juice having pH of 4.1.
 - **Answer:** (a) $7.94 \times 10^{-5} \text{ mol/L}$ & $1.26 \times 10^{-10} \text{ mol/L}$.
 - **Note:** $[\text{H}^+] = 10^{-4.1} = 7.94 \times 10^{-5} \text{ M}$. $[\text{OH}^-] = 10^{-14}/7.94 \times 10^{-5} = 1.26 \times 10^{-10} \text{ M}$.

- **Q:** Find the hydrogen ion concentration and the hydroxide ions concentration in **lemon juice** having a pH of 6.1. (Note: We had the tomato juice one, but this was a separate calculation!)
 - **A:** 7.94×10^{-7} mol/L & 1.26×10^{-8} mol/L.
 - | Note: $[\text{H}^+] = 10^{-6.1} = 7.94 \times 10^{-7} \text{ M}$.
 $[\text{OH}^-] = 10^{-14} / 7.94 \times 10^{-7} = 1.26 \times 10^{-8} \text{ M}$.
- **Q:** What is the rate constant (k) for the degradation of Hexane if its half-life is 15 days?
 - **A:** 0.0462 day^{-1} .
- **Q:** Calculate the exact mass of CO_2 released from burning 32 g of methane.
 - **A:** 88 g.
- **Q:** What is $[\text{H}^+]$ in a neutral solution? **A:** 10^{-7} mol/L, so pH = 7.
- **Q:** Below what pH should water be kept so $\geq 95\%$ of chlorine stays as HOCl? **A:** pH < 6.3.
- **Q:** What is the heat of reaction when methane is oxidized to CO_2 and liquid water? **A:** $-890.2 \text{ kJ/mol CH}_4$; it is negative because the reaction is exothermic.
- **Q:** How long does it take to remove 99% of CO from a room if $k = 1.2/\text{h}$? **A:** 3.8 hours ($t = \ln(100)/1.2$).
- **Q:** What makes a first-order half-life unique compared to zero-order? **A:** For first-order, $t_{1/2} = 0.693/k$, meaning it is constant and independent of the initial concentration.
- **Q:** What is a becquerel (Bq)? **A:** 1 radioactive disintegration per second.
- **Q:** What are the rate constants for Benzene, TCE, and Toluene? **A:** Benzene = 0.010/day; TCE = 0.003/day; Toluene = 0.058/day.

Slide-Specific Numerical Traps (From the Week-3 PDF)

- **Question (Ozone Photolysis):** What maximum wavelength of light would be capable of causing photolysis of ozone (O_3) into O_2 and O?
 - **Answer:** 1.13 μm .
 - | Note: Using $E = hc/\lambda$ and the enthalpy change of 104.6 kJ/mol, the max wavelength $\lambda \leq 1.13 \times 10^{-6} \text{ m}$.
- **Question (Fukushima Fish):** A fish caught off Fukushima has 25,000 Bq/kg of radioactive cesium (half-life = 3 years). What is the concentration after 5 years?
 - **Answer:** $\approx 7,875 \text{ Bq/kg}$.
 - | Note: Find $k = 0.693/3 = 0.231 \text{ yr}^{-1}$. Then use $C_t = C_0 e^{-kt} \rightarrow 25000 \times e^{-0.231(5)} \approx 7875 \text{ Bq/kg}$. It still vastly exceeds the safe limit of 1,200 Bq/kg.

- **Question (Hypochlorous Acid pH):** Below what pH should water be maintained so that at least 95% of chlorine is in the strong HOCl form rather than the weak OCl⁻ form?
 - **Answer:** Below pH 6.3.
 - *Note: Derived from the equilibrium constant $K = 10^{-7.60}$. If $[\text{HOCl}]/[\text{OCl}^-] = 19$, then $[\text{H}^+] = 10^{-6.32}$.*
- **Combustion Stoichiometry (Mass Calculations):**
 - **Methane (CH₄):** Equation: $\text{CH}_4 + 2\text{O}_2 \rightarrow \text{CO}_2 + 2\text{H}_2\text{O}$.
 - *Math:* 1 mol of CH₄ (16 g) produces 1 mol of CO₂ (44 g). Therefore, burning **32 g of methane** (2 mols) produces exactly **88 g of CO₂**.
 - **Butane (C₄H₁₀):** Equation: $2\text{C}_4\text{H}_{10} + 13\text{O}_2 \rightarrow 8\text{CO}_2 + 10\text{H}_2\text{O}$.
 - *Math:* 1 mol of Butane (58 g) produces 4 mols of CO₂ ($4 \times 44 = 176$ g). Therefore, burning **100 g of butane** produces exactly **303.4 g of CO₂**.
- **Half-Life (t_{1/2}) Specifics:** Formula: $k = 0.693/t_{1/2}$
 - **Xylene:** t_{1/2} = 56 days → $k = 0.693/56 = \mathbf{0.0124 \text{ day}^{-1}}$
 - **Hexane:** t_{1/2} = 15 days → $k = 0.693/15 = \mathbf{0.0462 \text{ day}^{-1}}$
- **Arithmetic Population Forecasting:** Formula: $P_n = P_0 + n\bar{x}$ (Where P₀ is the last known population, n is the number of decades into the future, and $\bar{x}$ is the average decadal increase).

ACTIVE RECALL & FEYNMAN CHALLENGE

Cover the answers below and test yourself before the exam.

Quiz 1: If you burn 58 grams of Butane (C₄H₁₀), how many grams of CO₂ will be produced?

Answer: 176 grams. (1 mole of Butane weighs 58g. The balanced equation shows 1 mole of Butane produces 4 moles of CO₂. $4 \times 44\text{g} = 176\text{g}$).

Quiz 2: If a chemical spill of Xylene has a half-life of 56 days, what is its decay rate constant (k)?

Answer: $k = 0.693/56 = 0.0124 \text{ day}^{-1}$.

Feynman Challenge: Explain the “10% Rule” (Ecological Efficiency) using a real-world example, and explain why food chains rarely have more than 4 or 5 levels.

Example Answer: Imagine a farmer grows 10,000 pounds of corn. When cows eat that corn, they burn most of the energy walking, mooing, and staying warm. Only

10% (1,000 pounds) of that energy actually turns into cow muscle (beef). If a human eats that beef, only 10% (100 pounds) turns into human mass. Because 90% of the energy is lost as heat at every single step, by the time you get to the 4th or 5th animal in the chain, there is simply no energy left to sustain another predator.